1. Description

1.1. Project

Project Name	foxilloscope
Board Name	NUCLEO-G474RE
Generated with:	STM32CubeMX 6.17.0
Date	07/23/2026

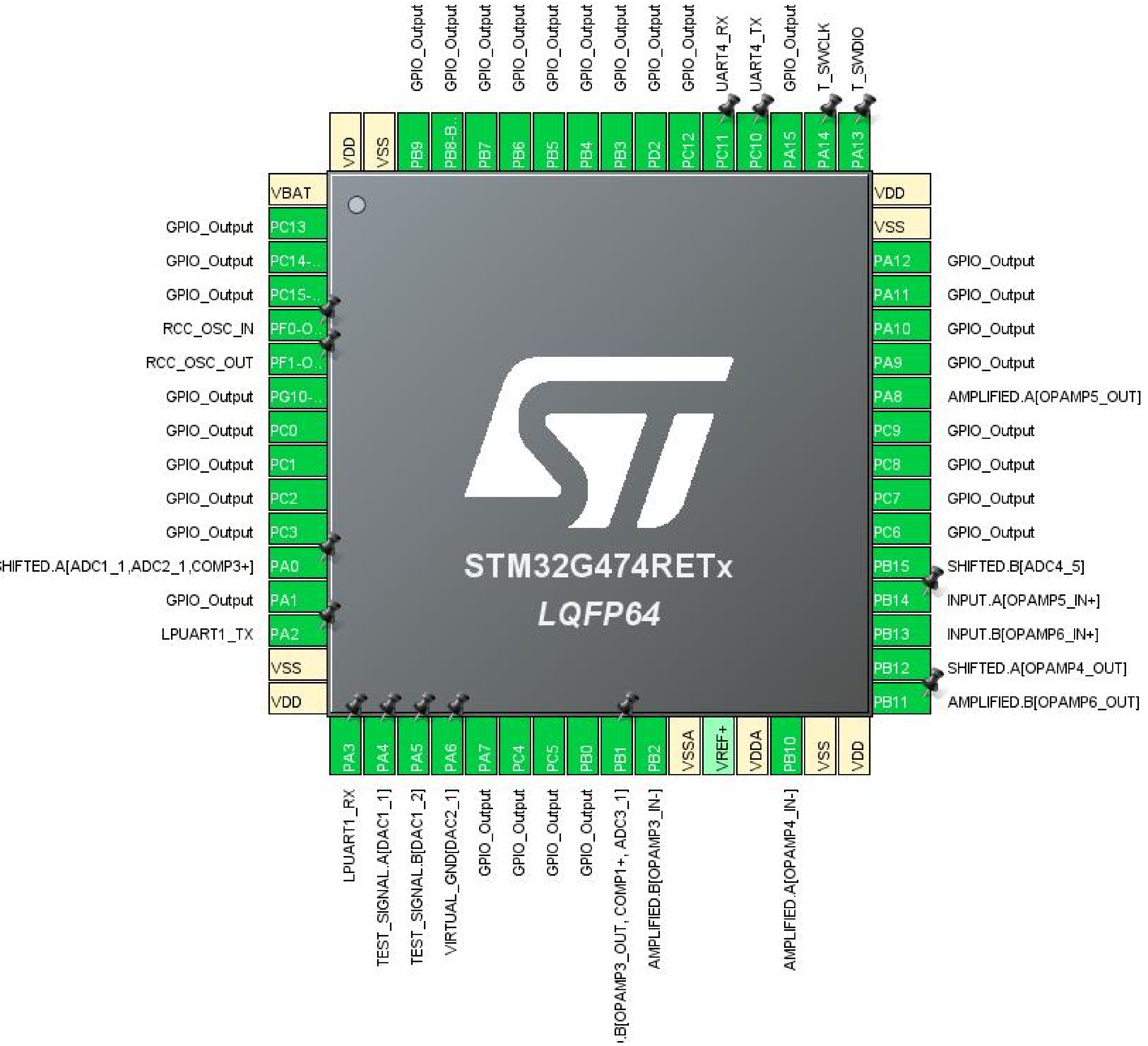
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4

2. Pinout Configuration



3. Pins Configuration

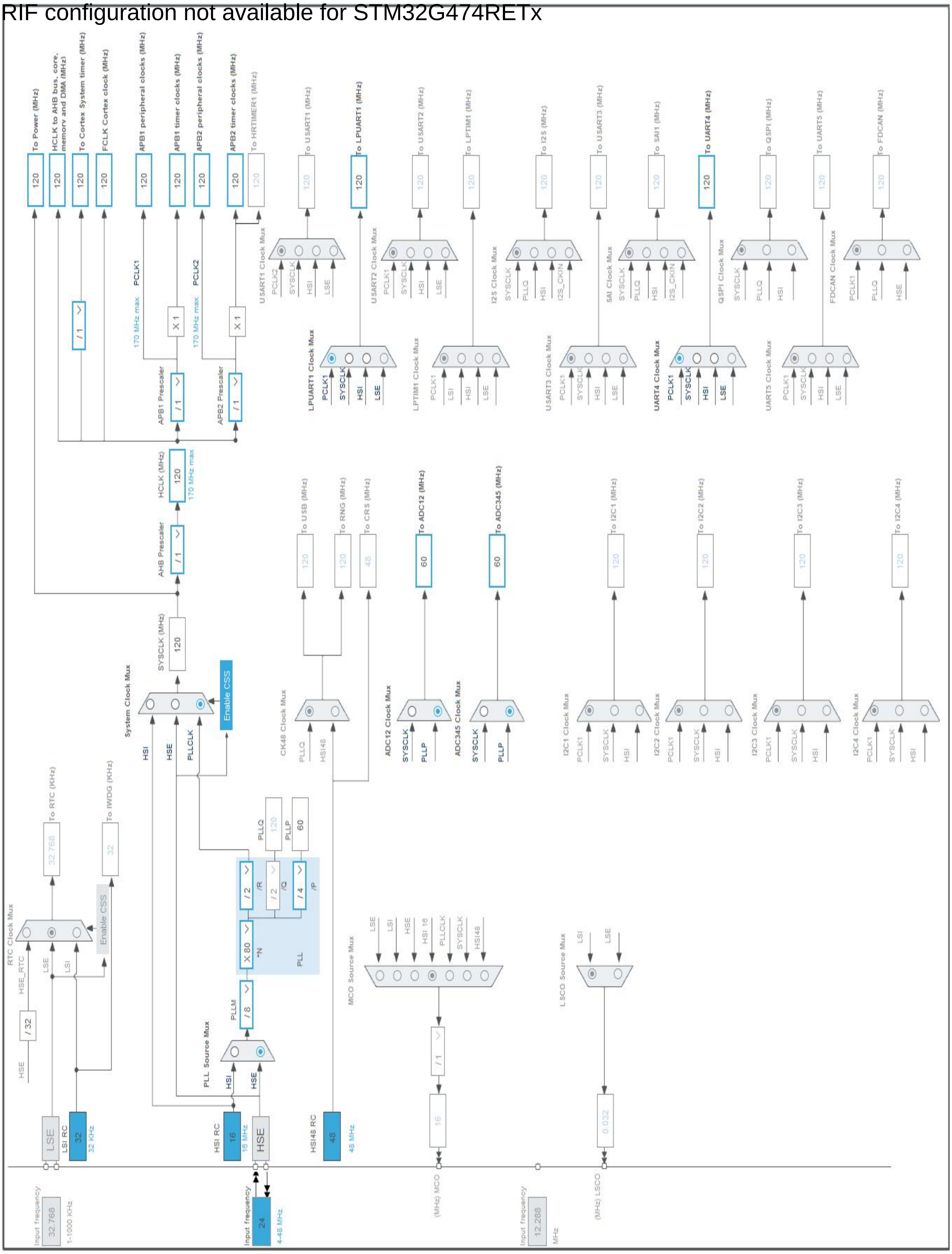
Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13 *	I/O	GPIO_Output	
3	PC14-OSC32_IN *	I/O	GPIO_Output	
4	PC15-OSC32_OUT *	I/O	GPIO_Output	
5	PF0-OSC_IN	I/O	RCC_OSC_IN	RCC_OSC_IN
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	RCC_OSC_OUT
7	PG10-NRST *	I/O	GPIO_Output	
8	PC0 *	I/O	GPIO_Output	
9	PC1 *	I/O	GPIO_Output	
10	PC2 *	I/O	GPIO_Output	
11	PC3 *	I/O	GPIO_Output	
12	PA0	I/O	ADC1_IN1, ADC2_IN1, COMP3_INP	SHIFTED.A[ADC1_1,ADC2_1,COMP3+]
13	PA1 *	I/O	GPIO_Output	
14	PA2	I/O	LPUART1_TX	
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	LPUART1_RX	
18	PA4	I/O	DAC1_OUT1	TEST_SIGNAL.A[DAC1_1]
19	PA5	I/O	DAC1_OUT2	TEST_SIGNAL.B[DAC1_2]
20	PA6	I/O	DAC2_OUT1	VIRTUAL_GND[DAC2_1]
21	PA7 *	I/O	GPIO_Output	
22	PC4 *	I/O	GPIO_Output	
23	PC5 *	I/O	GPIO_Output	
24	PB0 *	I/O	GPIO_Output	
25	PB1	I/O	OPAMP3_VOUT, COMP1_INP, ADC3_IN1	SHIFTED.B[OPAMP3_OUT, COMP1+, ADC3_1]
26	PB2	I/O	OPAMP3_VINM0	AMPLIFIED.B[OPAMP3_IN-]
27	VSSA	Power		
29	VDDA	Power		
30	PB10	I/O	OPAMP4_VINM0	AMPLIFIED.A[OPAMP4_IN-]
31	VSS	Power		
32	VDD	Power		
33	PB11	I/O	OPAMP6_VOUT	AMPLIFIED.B[OPAMP6_OUT]

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
34	PB12	I/O	OPAMP4_VOUT	SHIFTED.A[OPAMP4_OUT]
35	PB13	I/O	OPAMP6_VINP	INPUT.B[OPAMP6_IN+]
36	PB14	I/O	OPAMP5_VINP	INPUT.A[OPAMP5_IN+]
37	PB15	I/O	ADC4_IN5	SHIFTED.B[ADC4_5]
38	PC6 *	I/O	GPIO_Output	
39	PC7 *	I/O	GPIO_Output	
40	PC8 *	I/O	GPIO_Output	
41	PC9 *	I/O	GPIO_Output	
42	PA8	I/O	OPAMP5_VOUT	AMPLIFIED.A[OPAMP5_OUT]
43	PA9 *	I/O	GPIO_Output	
44	PA10 *	I/O	GPIO_Output	
45	PA11 *	I/O	GPIO_Output	
46	PA12 *	I/O	GPIO_Output	
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	T_SWDIO
50	PA14	I/O	SYS_JTCK-SWCLK	T_SWCLK
51	PA15 *	I/O	GPIO_Output	
52	PC10	I/O	UART4_TX	
53	PC11	I/O	UART4_RX	
54	PC12 *	I/O	GPIO_Output	
55	PD2 *	I/O	GPIO_Output	
56	PB3 *	I/O	GPIO_Output	
57	PB4 *	I/O	GPIO_Output	
58	PB5 *	I/O	GPIO_Output	
59	PB6 *	I/O	GPIO_Output	
60	PB7 *	I/O	GPIO_Output	
61	PB8-BOOT0 *	I/O	GPIO_Output	
62	PB9 *	I/O	GPIO_Output	
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32G474RETx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RETx
Datasheet	DS12288_Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

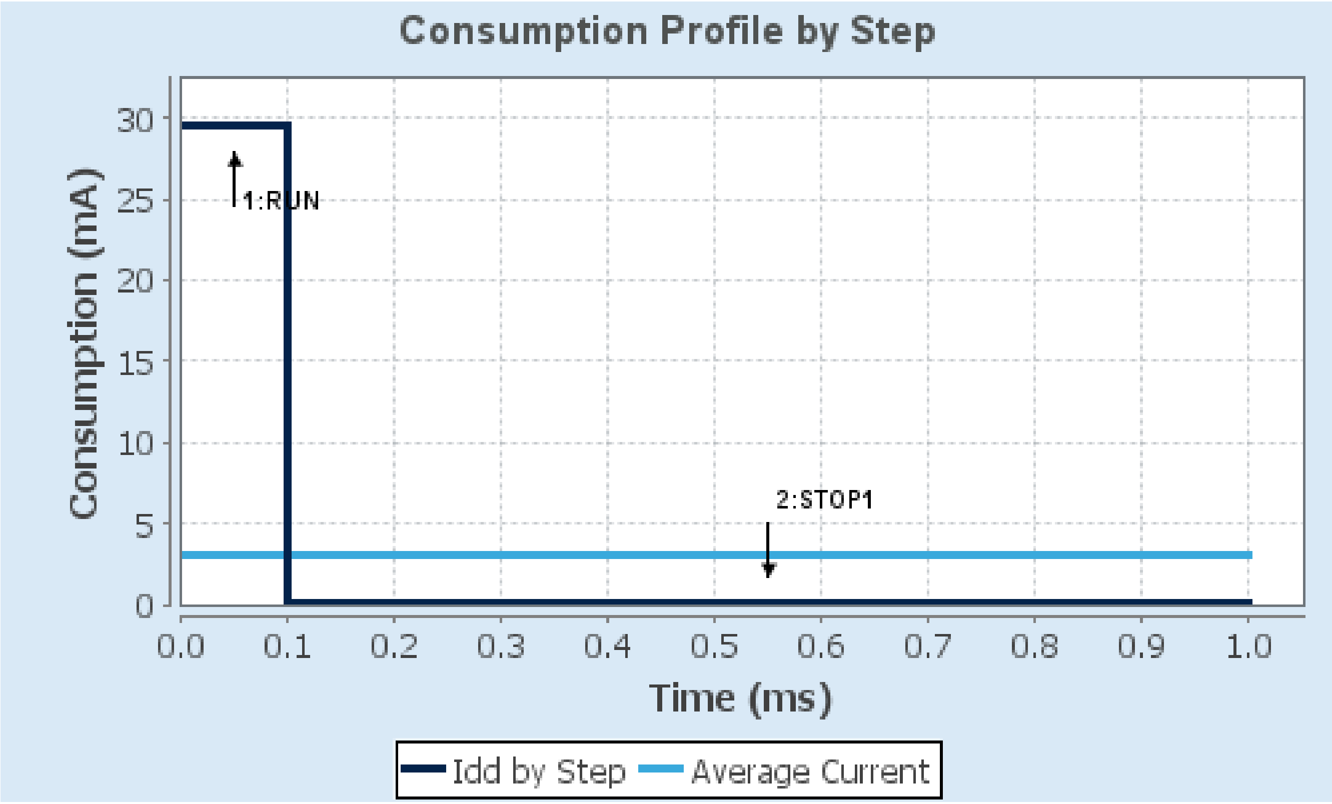
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 µA
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
30
25
20
15
10
5
0
1:RUN
2:STOP1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	foxilloscope
Project Folder	D:\projects\foxilloscope
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_DMA_Init	DMA
3	SystemClock_Config	RCC
4	MX_ADC3_Init	ADC3
5	MX_OPAMP3_Init	OPAMP3
6	MX_DAC1_Init	DAC1
7	MX_ADC4_Init	ADC4
8	MX_LPUART1_UART_Init	LPUART1
9	MX_TIM1_Init	TIM1
10	MX_TIM15_Init	TIM15
11	MX_TIM2_Init	TIM2

Rank	Function Name	Peripheral Instance Name
12	MX_DAC4_Init	DAC4
13	MX_OPAMP5_Init	OPAMP5
14	MX_DAC3_Init	DAC3
15	MX_ADC1_Init	ADC1
16	MX_ADC2_Init	ADC2
17	MX_OPAMP6_Init	OPAMP6
18	MX_OPAMP4_Init	OPAMP4
19	MX_COMP1_Init	COMP1
20	MX_COMP3_Init	COMP3
21	MX_DAC2_Init	DAC2
22	MX_UART4_Init	UART4

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:	
Mode	**Dual Interleaved mode + injected simultaneous mode ***
DMA Access Mode	DMA access mode enabled
Delay between 2 sampling phases	**7 Cycles ***
ADC_Settings:	
Clock Prescaler	**Asynchronous clock mode divided by 1 ***
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	**Enabled ***
Overrun behaviour	**Overrun data overwritten ***
ADC_Regular_ConversionMode:	
Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	**Timer 2 Trigger Out event ***
External Trigger Conversion Edge	Trigger detection on the rising edge
Rank	1
Channel	Channel 1
Sampling Time	2.5 Cycles
Offset Number	No offset
ADC_Injected_ConversionMode:	
Enable Injected Conversions	**Enable ***
Number Of Conversions	**1 ***
External Trigger Source	Injected Conversion launched by software
External Trigger Conversion Edge	None
Enable Injected Oversampling	Disable
Injected Conversion Mode	None

Injected Queue	Injected Queue Disable
Rank	1
Channel	**Channel Vrefint ***
Sampling Time	**640.5 Cycles ***
Offset Number	No offset

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.2. ADC2
IN1: IN1 Single-ended
3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	**Dual Interleaved mode + injected simultaneous mode ***
DMA Access Mode	DMA access mode enabled
Delay between 2 sampling phases	**7 Cycles ***

ADC_Settings:

Clock Prescaler	**Asynchronous clock mode divided by 1 ***
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
Rank	1
Channel	Channel 1
Sampling Time	2.5 Cycles

Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.3. ADC3
mode: IN1
3.3.1. Parameter Settings:

ADCs_Common_Settings:

Mode	**Dual interleaved mode only ***
DMA Access Mode	DMA access mode enabled
Delay between 2 sampling phases	**7 Cycles ***

ADC_Settings:

Clock Prescaler	**Asynchronous clock mode divided by 1 ***
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	**Enabled ***
Overrun behaviour	**Overrun data overwritten ***

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	**Timer 2 Trigger Out event ***
External Trigger Conversion Edge	Trigger detection on the rising edge
Rank	1
Channel	Channel 1

Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.4. ADC4
mode: IN5
3.4.1. Parameter Settings:

ADCs_Common_Settings:

Mode	**Dual interleaved mode only ***
DMA Access Mode	DMA access mode enabled
Delay between 2 sampling phases	**7 Cycles ***

ADC_Settings:

Clock Prescaler	**Asynchronous clock mode divided by 1 ***
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
Overrun behaviour	**Overrun data overwritten ***

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
Rank	1
Channel	Channel 5
Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.5. COMP1
mode: Input [+]
Input [-]: DAC3 OUT1
3.5.1. Parameter Settings:

Basic Parameters:

Trigger Mode	**Rising/Falling Edge Interrupt ***
Hysteresis Level	**Level 30mV ***

Output Configuration:

Blanking Source	None
Output Polarity	COMP output on GPIO isn't inverted

3.6. COMP3
mode: Input [+]
Input [-]: DAC3 OUT1
3.6.1. Parameter Settings:

Basic Parameters:

Trigger Mode	**Rising/Falling Edge Interrupt ***
Hysteresis Level	**Level 30mV ***

Output Configuration:

Blanking Source	None
Output Polarity	COMP output on GPIO isn't inverted

3.7. DAC1
OUT1 mode: Connected to external pin only
OUT2 mode: Connected to external pin only
3.7.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DAC High Frequency	Mode Automatic
DMA Double Data	**Enable ***
Signed Format	Disable
Trigger	**Timer 15 Trigger Out event ***
Trigger2	None
Wave generation mode	Disabled
User Trimming	Factory trimming

DAC Out2 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	**Timer 15 Trigger Out event ***
Trigger2	None
Wave generation mode	**Triangle wave generation ***
Maximum Triangle Amplitude	**2047 ***
User Trimming	Factory trimming

3.8. DAC2
OUT1 mode: Connected to external pin only
3.8.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None

Trigger2	None
User Trimming	Factory trimming

3.9. DAC3
mode: OUT1 mode
mode: OUT2 mode
3.9.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

DAC Out2 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

3.10. DAC4
mode: OUT1 mode
3.10.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None

Trigger2	None
User Trimming	Factory trimming

3.11. LPUART1
Mode: Asynchronous
3.11.1. Parameter Settings:

Basic Parameters:

Baud Rate	**460800 ***
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	**FIFO mode enable ***
Txfifo Threshold	**half full configuration ***
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.12. OPAMP3
Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS-DAC3_OUT2-INP
3.12.1. Parameter Settings:

Basic Parameters:

Power Mode	**High Speed ***
PGA Gain	**16 or -15 ***
User Trimming	Disable

3.13. OPAMP4
Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS-DAC4_OUT1-INP
3.13.1. Parameter Settings:

Basic Parameters:

Power Mode	**High Speed ***
PGA Gain	2 or -1
User Trimming	Disable

3.14. OPAMP5
Mode: Follower
3.14.1. Parameter Settings:

Basic Parameters:

Power Mode	**High Speed ***
User Trimming	Disable

3.15. OPAMP6
Mode: Follower
3.15.1. Parameter Settings:

Basic Parameters:

Power Mode	**High Speed ***
User Trimming	Disable

3.16. RCC
High Speed Clock (HSE): Crystal/Ceramic Resonator
3.16.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled

Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	3 WS (4 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE

3.17. SYS
Debug: Serial Wire
Timebase Source: TIM7
mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.18. TIM1
Slave Mode: External Clock Mode 1
Trigger Source: ITR1
Channel1: PWM Generation No Output
mode: One Pulse Mode

3.18.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable
Slave Mode Controller	ETR mode 1

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	**Output Compare (OC1REF) ***
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	**99 ***
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	**Set ***

3.19. TIM2
Slave Mode: Gated Mode
Trigger Source: ITR0
Clock Source : Internal Clock
3.19.1. Parameter Settings:

Counter Settings:	
Prescaler (PSC - 16 bits value)	**39 ***
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	**19 ***
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable
Slave Mode Controller	Gated Mode
Trigger Output (TRGO) Parameters:	
Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	**Update Event ***

3.20. TIM15
mode: Clock Source
3.20.1. Parameter Settings:

Counter Settings:	
Prescaler (PSC - 16 bits value)	**30 ***
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	**16 ***
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable
Trigger Output (TRGO) Parameters:	
Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	**Update Event ***

3.21. UART4
Mode: Asynchronous
3.21.1. Parameter Settings:

Basic Parameters:	
Baud Rate	**460800 ***
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1
Advanced Parameters:	
Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	**FIFO mode enable ***
Txfifo Threshold	**half full configuration ***
Rxfifo Threshold	1 eighth full configuration
Advanced Features:	
Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.22. FREERTOS
Interface: CMSIS_V2
3.22.1. Config parameters:

API:	
FreeRTOS API	CMSIS v2
Versions:	
FreeRTOS version	10.3.1
CMSIS-RTOS version	2.1.3
MPU/FPU:	
ENABLE_MPU	Disabled

ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
USE_SB_COMPLETED_CALLBACK	0
USE_MINI_LIST_ITEM	1
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	**30720 ***
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	**Enabled ***
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	**Enabled ***
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled

TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.22.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.22.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled

Project settings (see parameter description first):

Use FW pack heap file	Enabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.A[ADC1_1,ADC2_1,COMP3+]
ADC2	PA0	ADC2_IN1	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.A[ADC1_1,ADC2_1,COMP3+]
ADC3	PB1	ADC3_IN1	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.B[OPAMP3_OUT, COMP1+, ADC3_1]
ADC4	PB15	ADC4_IN5	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.B[ADC4_5]
COMP1	PB1	COMP1_INP	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.B[OPAMP3_OUT, COMP1+, ADC3_1]
COMP3	PA0	COMP3_INP	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.A[ADC1_1,ADC2_1,COMP3+]
DAC1	PA4	DAC1_OUT1	Analog mode	No pull-up and no pull-down	n/a	TEST_SIGNAL.A[DAC1_1]
	PA5	DAC1_OUT2	Analog mode	No pull-up and no pull-down	n/a	TEST_SIGNAL.B[DAC1_2]
DAC2	PA6	DAC2_OUT1	Analog mode	No pull-up and no pull-down	n/a	VIRTUAL_GND[DAC2_1]
LPUART1	PA2	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	LPUART1_RX	Alternate Function Push Pull	**Pull-down ***	Low	
OPAMP3	PB1	OPAMP3_VOUT	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.B[OPAMP3_OUT, COMP1+, ADC3_1]
	PB2	OPAMP3_VINM0	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.B[OPAMP3_IN-]
OPAMP4	PB10	OPAMP4_VINM0	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.A[OPAMP4_IN-]
	PB12	OPAMP4_VOUT	Analog mode	No pull-up and no pull-down	n/a	SHIFTED.A[OPAMP4_OUT]
OPAMP5	PB14	OPAMP5_VINP	Analog mode	No pull-up and no pull-down	n/a	INPUT.A[OPAMP5_IN+]
	PA8	OPAMP5_VOUT	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.A[OPAMP5_OUT]
OPAMP6	PB11	OPAMP6_VOUT	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.B[OPAMP6_OUT]
	PB13	OPAMP6_VINP	Analog mode	No pull-up and no pull-down	n/a	INPUT.B[OPAMP6_IN+]
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	RCC_OSC_IN
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	RCC_OSC_OUT
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	T_SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	T_SWCLK
UART4	PC10	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC11	UART4_RX	Alternate Function Push Pull	**Pull-down ***	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
GPIO	PC13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC14-OSC32_IN	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC15-OSC32_OUT	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PG10-NRST	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PA1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PA7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PA9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PA10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PA11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PA12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PA15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB8-BOOT0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	

4.2. DMA configuration

DMA request	Stream	Direction	Priority
MEMTOMEM	DMA1_Channel2	Memory To Memory	Low
LPUART1_TX	DMA1_Channel4	Memory To Peripheral	Low
ADC1	DMA1_Channel5	Peripheral To Memory	Low
MEMTOMEM	DMA1_Channel6	Memory To Memory	Low
DAC1_CH1	DMA1_Channel1	Memory To Peripheral	**Very High ***
ADC3	DMA1_Channel3	Peripheral To Memory	Low
UART4_TX	DMA1_Channel7	Memory To Peripheral	Low

MEMTOMEM: DMA1_Channel2 DMA request Settings:

Mode: Normal
Src Memory Increment: **Enable ***
Dst Memormy Increment: **Enable ***
Src Memory Data Width: **Word ***
Dst Memormy Data Width: **Word ***

LPUART1_TX: DMA1_Channel4 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

ADC1: DMA1_Channel5 DMA request Settings:

Mode: **Circular ***
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: **Word ***
Memory Data Width: **Word ***

MEMTOMEM: DMA1_Channel6 DMA request Settings:

Mode: Normal

Src Memory Increment:	**Enable ***
Dst Memormy Increment:	**Enable ***
Src Memory Data Width:	**Word ***
Dst Memormy Data Width:	**Word ***

DAC1_CH1: DMA1_Channel1 DMA request Settings:

Mode:	**Circular ***
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Word
Memory Data Width:	**Word ***

ADC3: DMA1_Channel3 DMA request Settings:

Mode:	**Circular ***
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	**Word ***
Memory Data Width:	**Word ***

UART4_TX: DMA1_Channel7 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
DMA1 channel1 global interrupt	true	5	0
DMA1 channel2 global interrupt	true	5	0
DMA1 channel3 global interrupt	true	5	0
DMA1 channel4 global interrupt	true	5	0
DMA1 channel5 global interrupt	true	5	0
DMA1 channel7 global interrupt	true	5	0
TIM1 capture compare interrupt	true	5	0
UART4 global interrupt / UART4 wake-up interrupt through EXTI line 34	true	5	0
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	true	15	0
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	true	0	0
LPUART1 global interrupt	true	5	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
DMA1 channel6 global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
TIM1 break interrupt and TIM15 global interrupt	unused		
TIM1 update interrupt and TIM16 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM2 global interrupt	unused		
ADC3 global interrupt	unused		
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	unused		
ADC4 global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
FPU global interrupt	unused		

4.3.2. NVIC Code generation

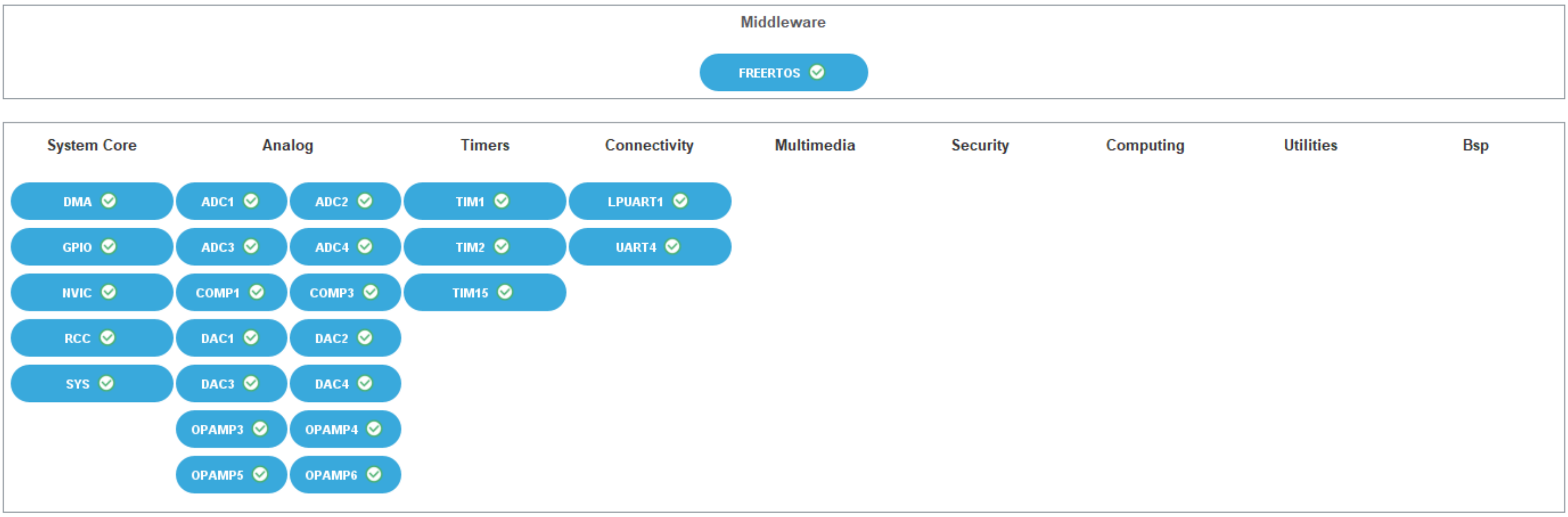
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
DMA1 channel1 global interrupt	false	true	true
DMA1 channel2 global interrupt	false	true	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
DMA1 channel7 global interrupt	false	true	true
TIM1 capture compare interrupt	false	true	true
UART4 global interrupt / UART4 wake-up interrupt through EXTI line 34	false	true	true
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	false	true	true
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	false	true	true
LPUART1 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32c5.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

	guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5310-guideline-for-using-analog-features-of-stm32g4-series-versus-stm32f3-series-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5346-stm32g4-adc-

	use-tips-and-recommendations-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5788-stm32-digital-power-pid-and-iir-filters-for-smps-control-design-and-comparison-on-bg414edpow1-discovery-kit-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4894-how-to-use-eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5978-introduction-to-mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2548-introduction-to-dma-controller-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4566-how-to-extend-

	the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-how-to-use-dac-for-audio-and-waveform-generation-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-how-to-use-usart-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-how-to-use-usb-dfu-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-how-to-use-spi-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4760-introduction-to-quadspi-interface-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-how-to-use-generalpurpose-timer-peripheral-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-introduction-to-debug-toolbox-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5094-migrating-between-stm32f334303-and-stm32g431g474g491-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5738-guidelines-for-lifetime-estimation-on-stm32g4-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6363-introduction-to-clock-requirements-and-calibration-for-stm32-mcus-stmicroelectronics.pdf

Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0430-stm32g471xx473xx474xx483xx484xx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00431551.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0440-stm32g4-series-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-devices-in-bga-packages-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf